

BUSINESS INTELLIGENCE AND TOOLS

(24CSH-341)

EXPERIMENT 4: OLAP Operations (Slice, Dice, Drill-up, and Drill-down) using Tableau

AIM To design and implement OLAP operations using Microsoft Power BI on the Global Superstore dataset, enabling interactive analysis of business data through Slicing, Dicing, Drill Down, and Drill Up operations for effective reporting and decision-making.

OBJECTIVES

- Understand OLAP and its role in Business Intelligence.
 - Perform Slicing and Dicing using filters.
 - Implement Drill Down and Drill Up using hierarchies.
 - Analyze Sales, Profit, Orders, Category, Region, and Year.
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TOOLS & SOFTWARE REQUIRED

- Tableau Desktop / Tableau Public
 - Global Superstore Dataset.xlsx
 - Tableau Worksheets and Dashboard
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THEORY

OLAP (Online Analytical Processing) is a technology used in Business Intelligence (BI) to analyze multidimensional business data and support effective decision-making. It allows users to examine data from different perspectives, identify trends, compare performance, and generate meaningful reports.

In this experiment, Tableau is used to perform OLAP operations on the Global Superstore dataset. The dataset contains information such as Sales, Profit, Orders, Category, Segment, Region, and Year. Tableau provides interactive worksheets, filters, charts, and dashboards for analyzing this data.

1. **Slicing:** Slicing is an OLAP operation that selects a particular value or subset from one dimension of data. In Tableau, it is performed using filters. For example, selecting a single Region displays the sales and profit information for that region only.
2. **Dicing:** Dicing is an OLAP operation that selects a subset of data using multiple dimensions simultaneously. In Tableau, filters such as Region, Category, and Year can be applied together to analyze a specific combination of business data.
3. **Drill Down:** Drill Down is the process of moving from summarized data to more detailed information. In Tableau, it can be performed using hierarchies. For example, geographical data can be analyzed from Region → Country → City → State.
4. **Drill Up:** Drill Up is the reverse of Drill Down. It moves from detailed data to higher-level summaries. In Tableau, users can collapse hierarchies to analyze data from State → City → Country → Region.

OLAP operations help organizations analyze business performance at different levels of detail. Tableau dashboards make this analysis interactive, allowing users to apply filters, explore geographical data, compare sales and profit, and identify useful business insights for reporting and decision-making.

PROCEDURE

Part A: Load Dataset

1. Open Tableau.
2. Go to Connect → Microsoft Excel.
3. Select **Global Superstore Dataset.xlsx**.
4. Open the required worksheet and verify fields such as Sales, Profit, Order ID, Region, Country, City, State, Category, and Year.

Part B: Create Worksheets

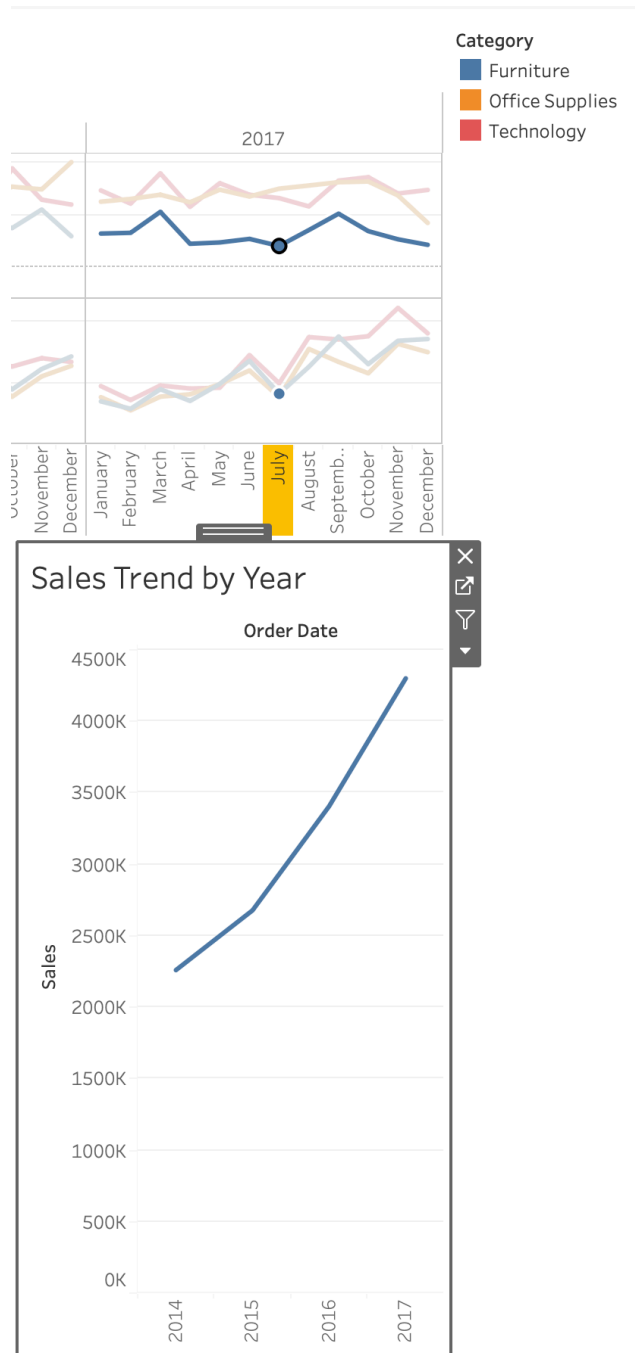
Create the following sheets:

Worksheet	Visualization
Sales KPI	Total Sales
Profit KPI	Total Profit
Orders KPI	Count of Orders
Sales by Category	Bar chart
Sales by Region	Bar chart
Sales Trend by Year	Line chart

Part C: OLAP Slicing

1. Create a worksheet named OLAP – Slicing.
2. Drag **Region** to the Filters shelf.
3. Select one Region.
4. Observe the changes in Sales, Profit, and charts.

Part D: OLAP Dicing



1. Create a worksheet named OLAP – Dicing.
2. Add filters for **Region**, **Category**, and **Year**.
3. Select multiple values.
4. Analyze the selected subset using charts and KPIs.

Part E: OLAP Drill Down

1. Create a geographical hierarchy:

Region → **Country** → **City** → **State**

2. Drag the hierarchy to Rows or Columns.
3. Click the + (Expand) icon to drill down.
4. Analyze sales and profit at different geographical levels.

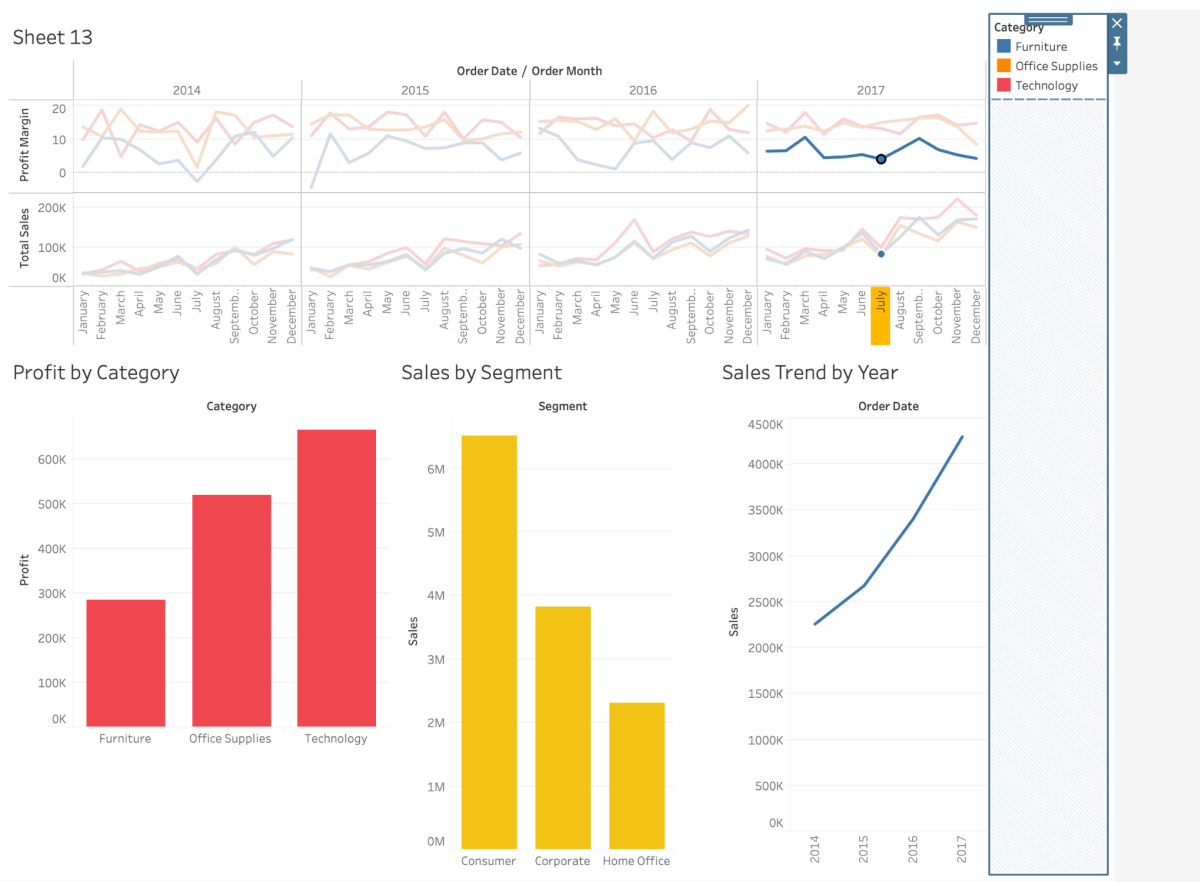
Part F: OLAP Drill Up

1. Use the same geographical hierarchy.
2. Start from State or City level.
3. Click the – (Collapse) icon to drill up.
4. Observe summarized sales and profit at higher levels.

Part G: Dashboard and Validation

1. Create a dashboard by combining the worksheets.
2. Add filters and enable interactions.
3. Verify that Slicing, Dicing, Drill Down, and Drill Up work correctly.
4. Compare results at different levels of detail.

Sheet 13



RESULT

OLAP operations were successfully implemented in Tableau using the Global Superstore dataset. Filters and geographical hierarchies enabled interactive analysis of Sales, Profit, Orders, Category, Region, and Year for effective business reporting and decision-making.

LEARNING OUTCOMES

1. Understood the concept of OLAP and its role in Business Intelligence.
2. Learned to perform multidimensional data analysis using Tableau.
3. Learned to implement Slicing using filters.
4. Learned to implement Dicing using multiple dimensional filters.
5. Learned to create and use hierarchies for OLAP analysis.
6. Learned to perform Drill Down from summarized data to detailed data.
7. Learned to perform Drill Up from detailed data to summarized data.
8. Gained practical experience in creating interactive Tableau dashboards for business reporting and decision-making.